Data & AI

15. How an LLM Answers

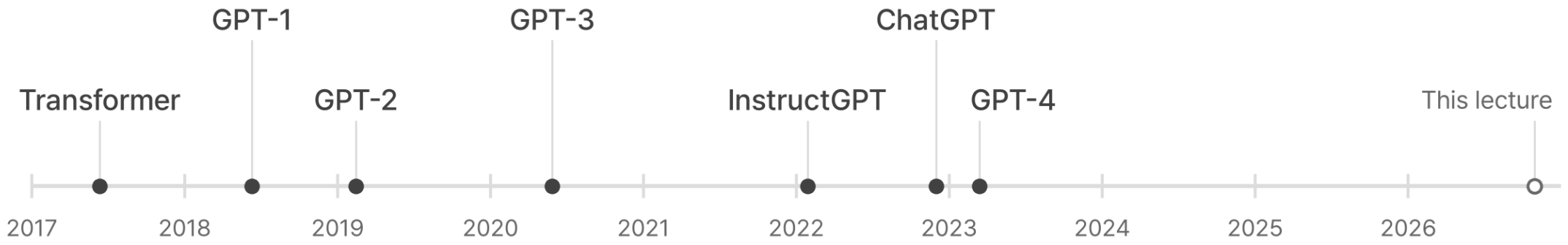
Dasaem Jeong · Art & Technology, Sogang University

Assignment 2 Mid-point

jdasam.github.io/mas1004-leaderboard

- The leaderboard this morning
- A few of you: one change you made, and what it did to your validation score
- Tuesday 10 November, 22:00: Assignment 2 due

Last Time



- Every model from GPT-1 to GPT-4 did one thing: predict the next word
- What changed: the training text, the size, and human feedback

Today

- Where an answer comes from: the parameters, the context, and tools
- Reasoning models: writing the steps before the answer

Where an Answer Comes From

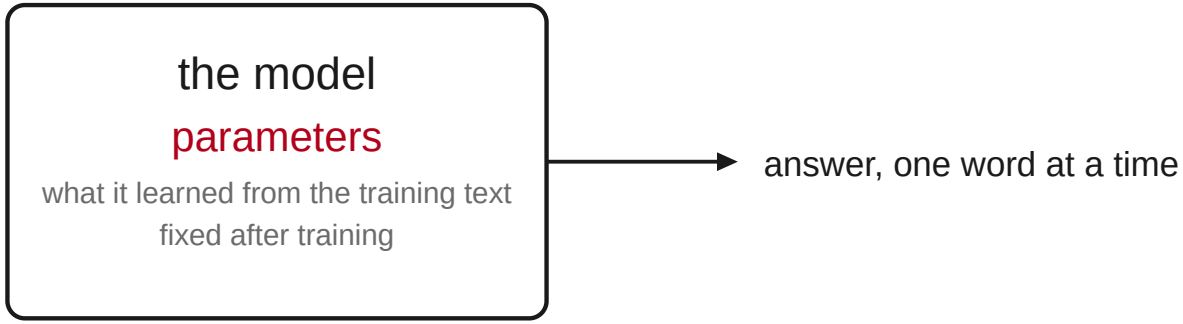
Two Questions

> At what temperature does water boil?

> Who hit the game-winning hit for SK against Hanwha in Daejeon on 7 June 2009?

- Both facts are old and public. Try each with web search off, then on.

Answers from the Parameters



- Nothing is looked up: the answer is whatever the training text made likely

Closed-Book Question Answering

Trivia questions, with nothing but the question as input

TriviaQA questions, in the format GPT-3 was tested with

> Q: 'Nude Descending A Staircase' is perhaps the most famous painting by which 20th century artist?
> A: Marcel Duchamp

> Q: What is solid CO2 commonly called?
> A: dry ice

> Q: Who wrote the novel The Eagle Has Landed?
> A: Jack Higgins

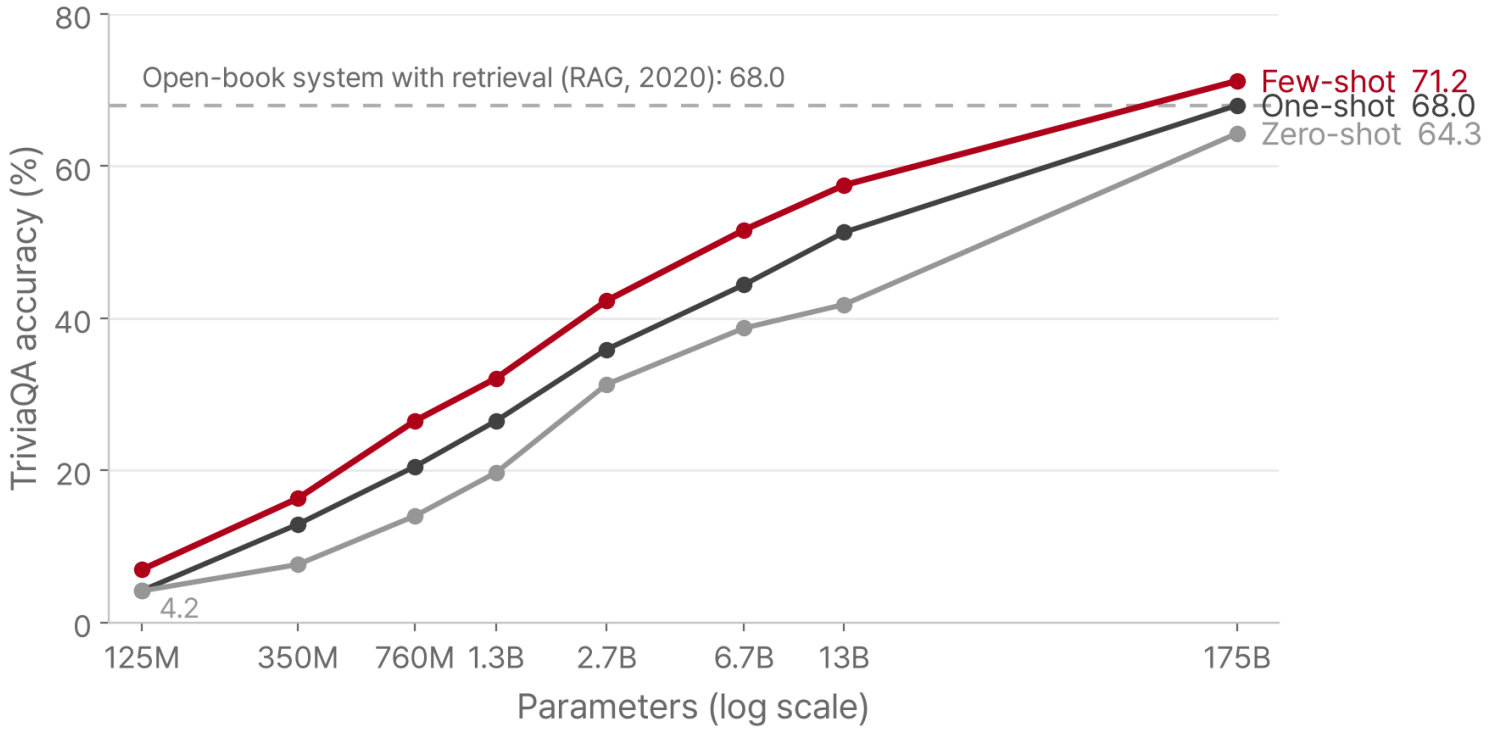
> Q: The Dodecanese Campaign of WWII that was an attempt by the Allied forces to capture islands in the Aegean Sea was the inspiration for which acclaimed 1961 commando film?
> A: The Guns of Navarone

- No document, no search: the answer has to be in the parameters already

Joshi et al. (2017), TriviaQA · Brown et al. (2020), Figure G.34 · "closed-book": Roberts, Raffel, Shazeer (2020)

TriviaQA Accuracy and Model Size

Brown et al. (2020), Language Models are Few-Shot Learners: eight sizes of GPT-3



- 125M parameters: 4 out of 100 right. 175B parameters: 64 out of 100

Brown et al. (2020), Table H.1 (dev set) and Table 3.3 · T5 sizes in Roberts et al. (2020), Table 1

Small and Large Models on SimpleQA

OpenAI (2024), SimpleQA: 4,326 short fact questions, one indisputable answer each

> Who received the IEEE Frank Rosenblatt Award in 2010? Michio Sugeno

> What day, month, and year was Carrie Underwood's album "Cry Pretty" certified Gold by the RIAA?
> October 23, 2018

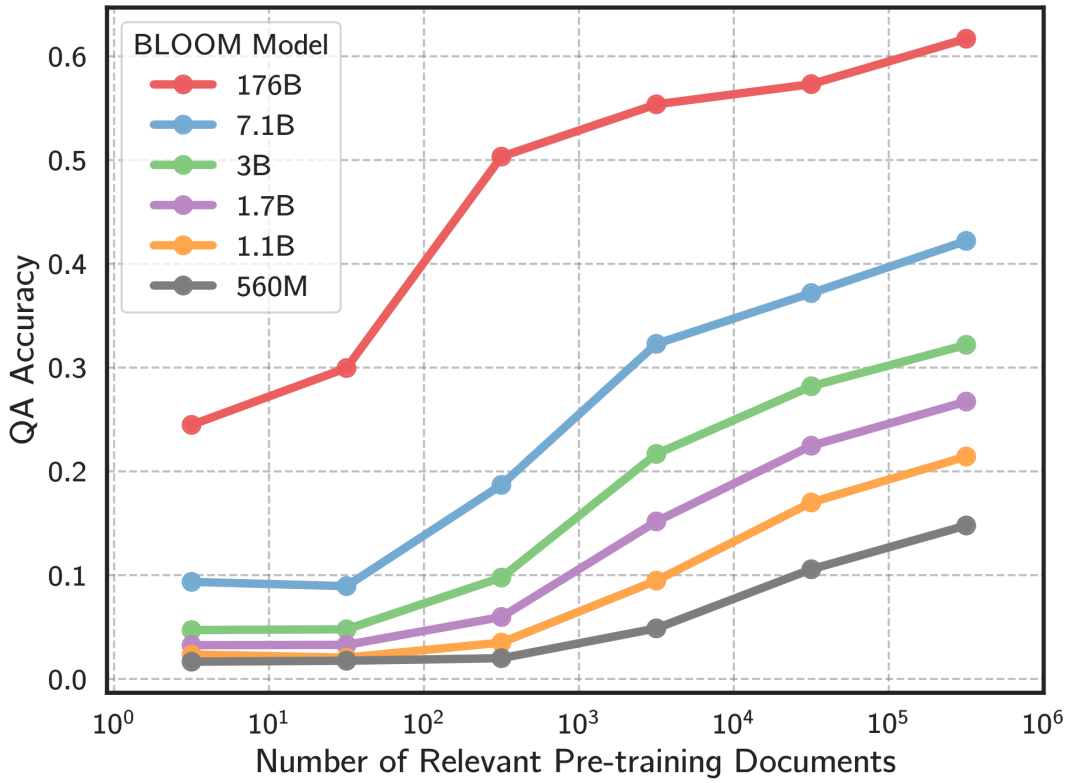
model	correct	not attempted	incorrect
Claude 3 Haiku	5.1%	75.3%	19.6%
Claude 3 Opus	23.5%	39.6%	36.9%
GPT-4o mini	8.6%	0.9%	90.5%
GPT-4o	38.2%	1.0%	60.8%

- The small model of each family gets far fewer right. One abstains, one guesses

Wei et al. (2024), Measuring short-form factuality in large language models, Table 1 and Table 3

Long-Tail Knowledge

Kandpal et al. (2023), Large Language Models Struggle to Learn Long-Tail Knowledge, ICML 2023

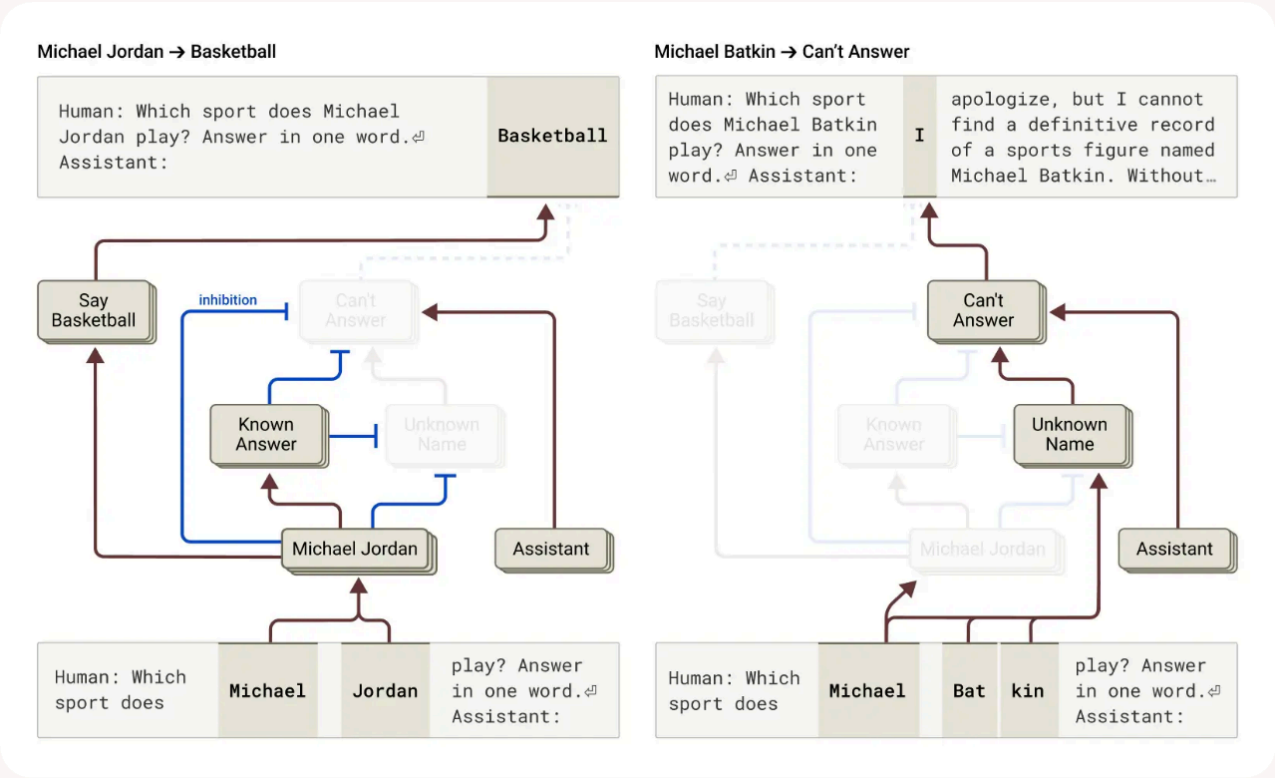


- The fewer training documents mention a fact, the less often the model answers correctly

Kandpal, Deng, Roberts, Wallace, Raffel (2023), Figure 1

A Known Name and an Unknown Name

Anthropic (2025), Tracing the thoughts of a large language model



- The default is to decline; a "known entity" signal switches the default off
- A hallucination: the signal fires for a name the model only half knows

Why Language Models Hallucinate

OpenAI (2025) · short factual questions (SimpleQA)

model	abstains	correct	wrong
gpt-5-thinking-mini	52%	22%	26%
o4-mini	1%	24%	75%

- On a leaderboard that counts only correct answers, the second model wins
- Like a test with no penalty for wrong answers: guessing beats leaving it blank

Fake Cases in a Court Filing

United States District Court, S.D.N.Y., Opinion and Order on Sanctions (22 June 2023)

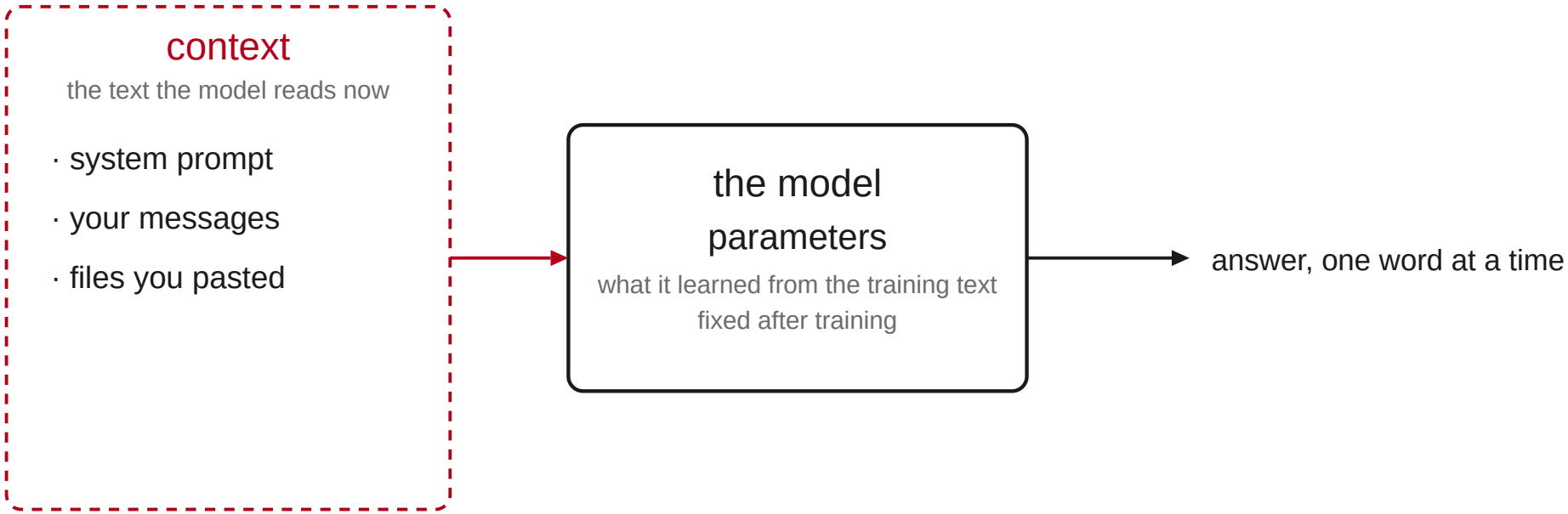
> The lawyer, under oath: "I heard about this new site which I assumed -- I falsely assumed was like a super search engine called ChatGPT"

> He had asked ChatGPT: "Is Varghese a real case" and "Are the other cases you provided fake"

> "ChatGPT responded that it had supplied “real” authorities that could be found through Westlaw, LexisNexis and the Federal Reporter."

- The cited court decisions did not exist; a $5,000 penalty for filing them and standing by them

Answers from the Context



- The context is the text the model reads right now: it is read, not learned

The System Prompt

Claude Sonnet 5.5 on claude.ai, the version of 27 October 2026, one paragraph of many

```
<knowledge_cutoff>
Claude's reliable knowledge cutoff, past which it can't answer reliably, is the end of Jun 2026. It answers
the way a highly informed individual in Jun 2026 would if talking to someone from {{currentDateTime}}, and
can say so when relevant. For events or news that may post-date the cutoff, Claude often can't know either
way and says so. For current news or events (e.g. current officeholders), Claude gives its most recent pre-
cutoff information, notes it may be outdated, and points to web search. [...]
</knowledge_cutoff>
```

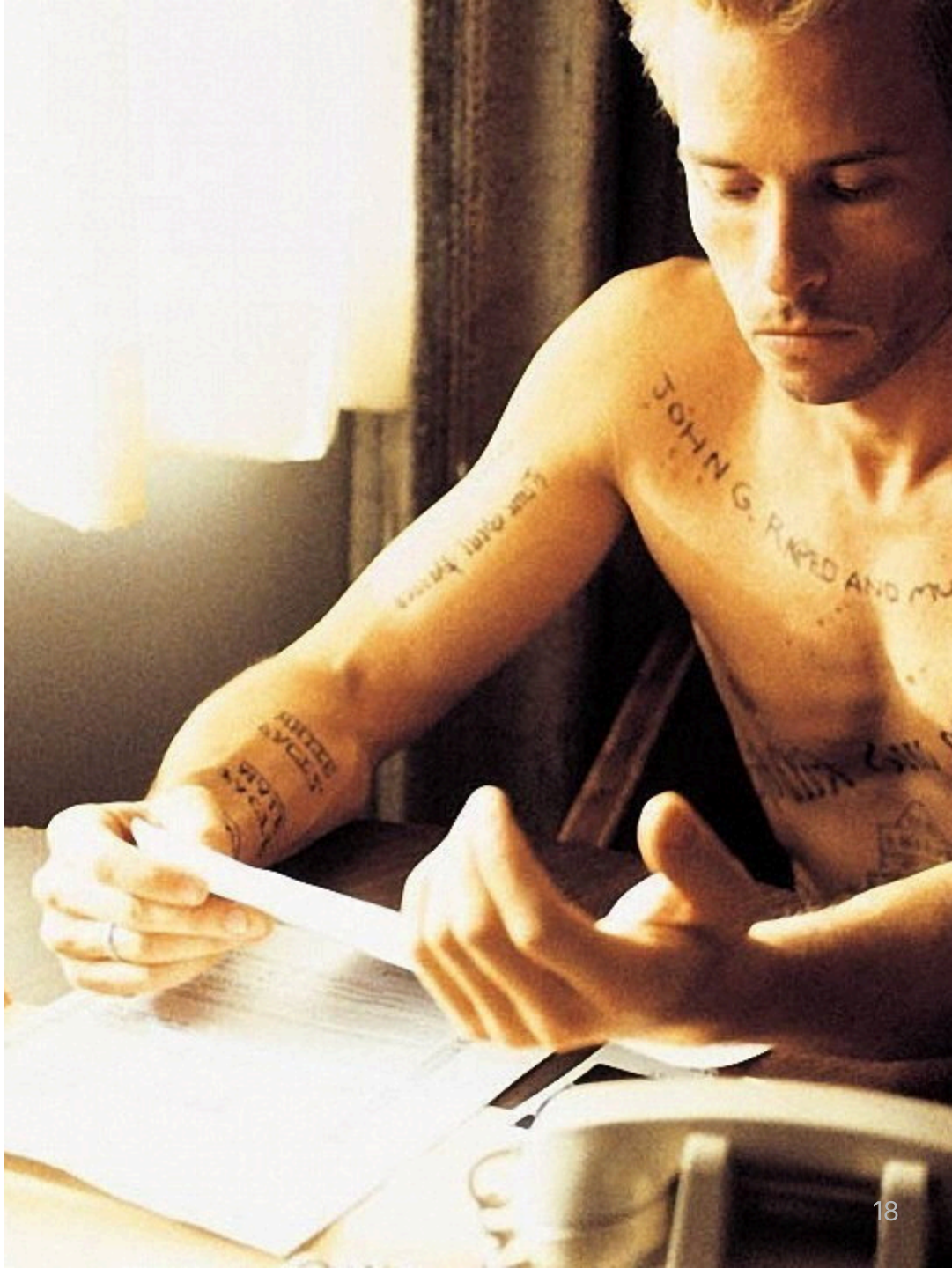
- The product writes today's date and the cutoff into the context, as text the model reads

platform.claude.com/docs/en/release-notes/system-prompts/claude-sonnet-5-5



Memento and an LLM

the film	the LLM
what Leonard knew before the accident	parameters
the photos and notes in his hands	context
tattoos, the stack of photos, files	memory and tools outside the model



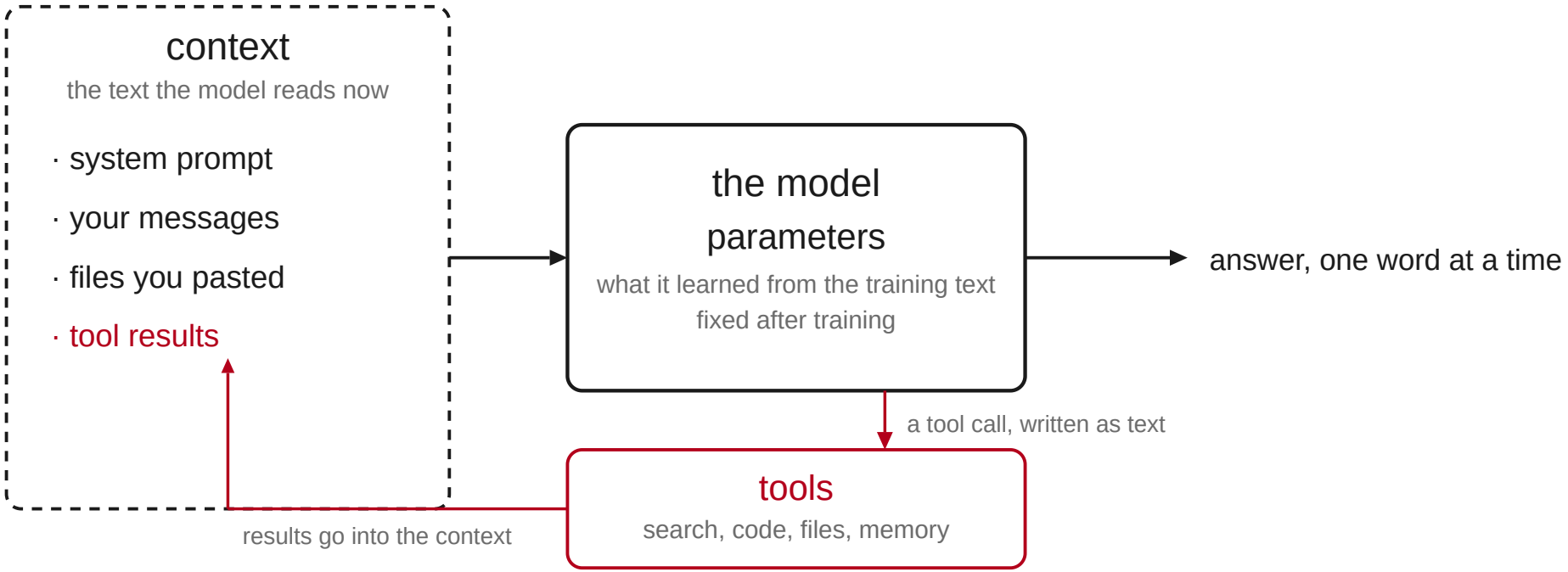
Memory in ChatGPT and Claude

"It remembers me"

- Nothing about you enters the parameters: the weights never change while you chat
- The product stores notes as text outside the model
- In a new chat, the relevant notes are put back into the context

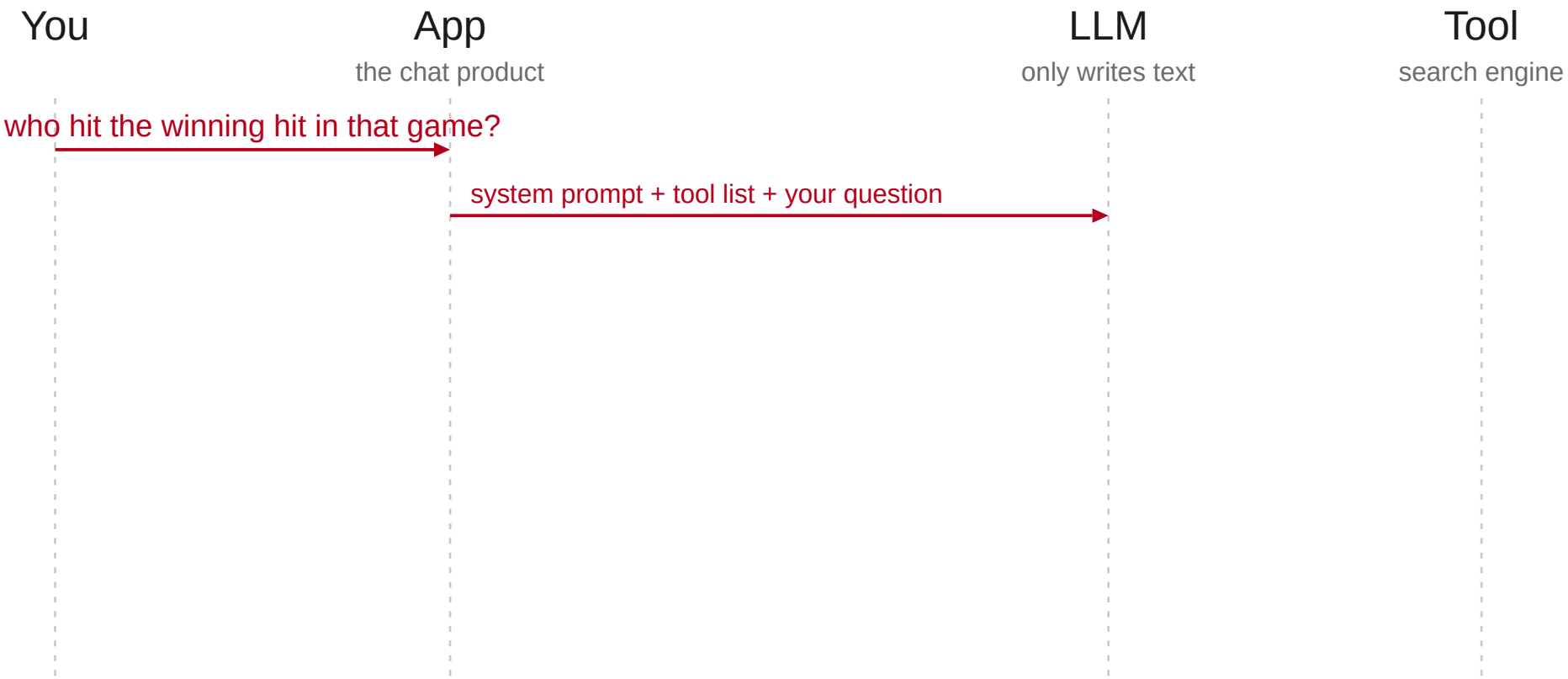
OpenAI, Memory FAQ · Anthropic (2025-09-11), Bringing memory to Claude

Answers from Tools

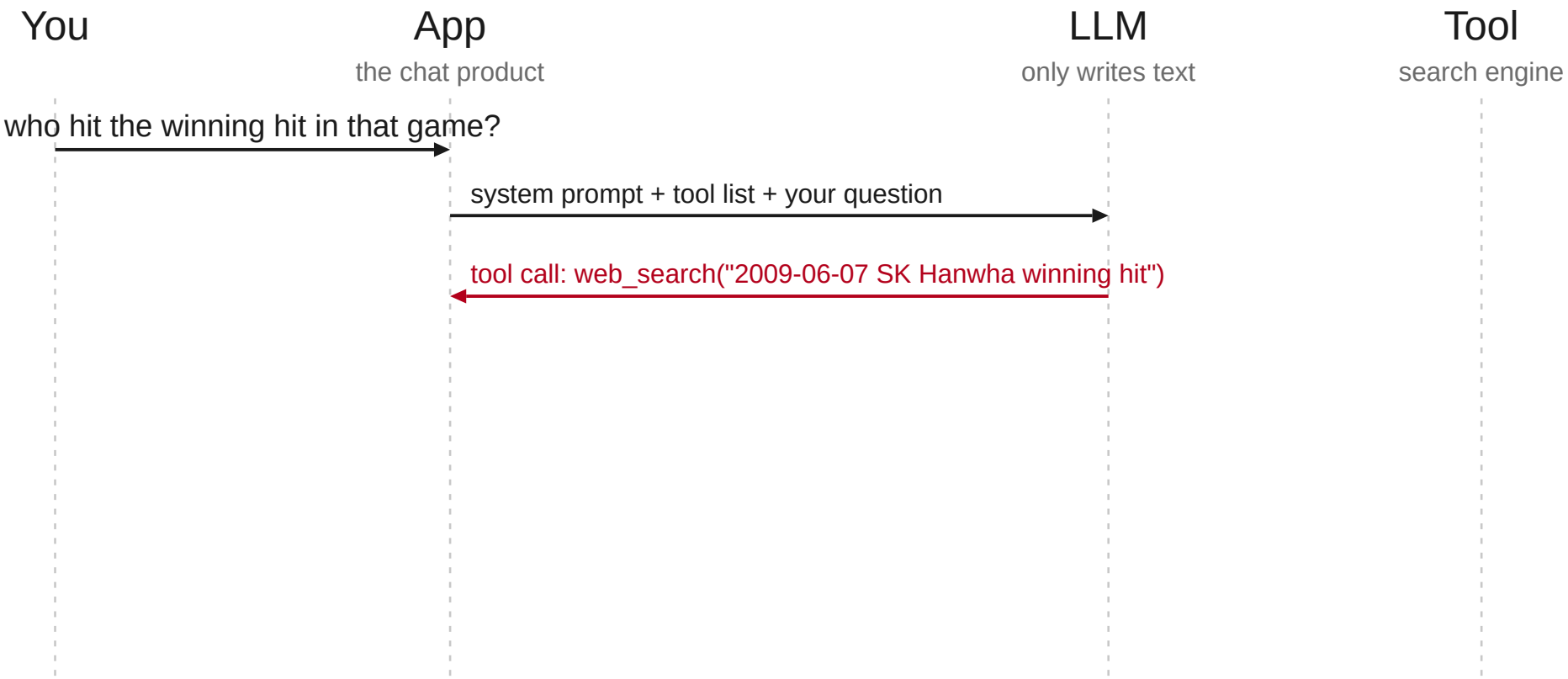


- The model never searches: it writes a request, a program runs it, the result is pasted into the context

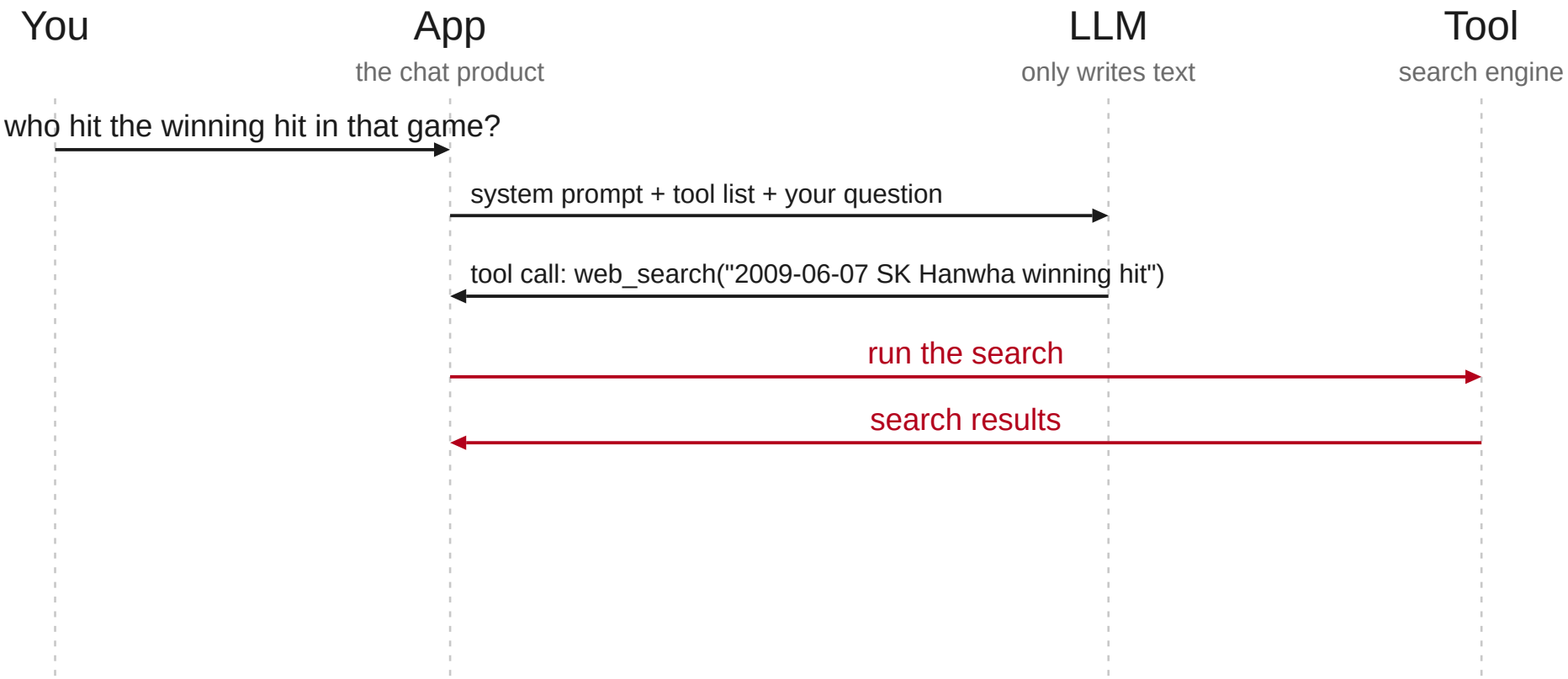
When the App Searches



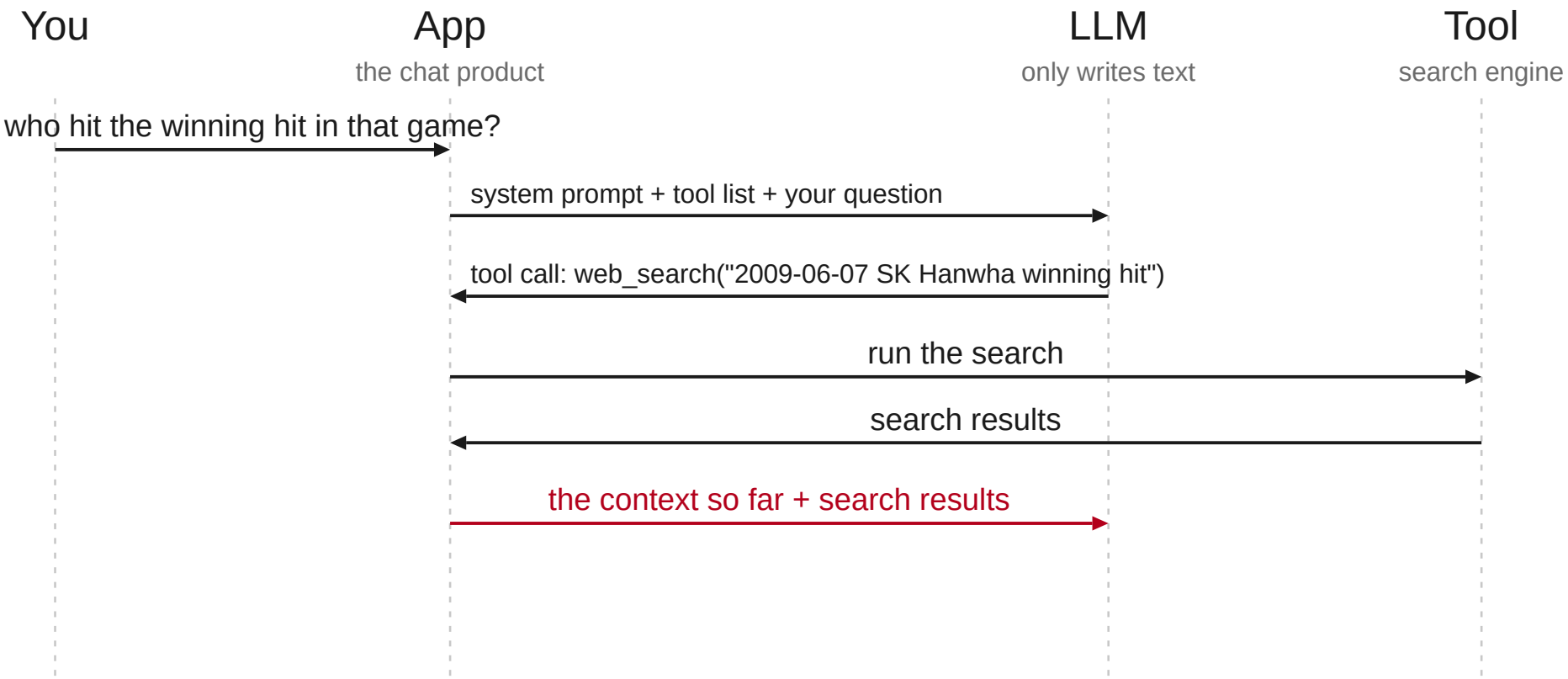
When the App Searches



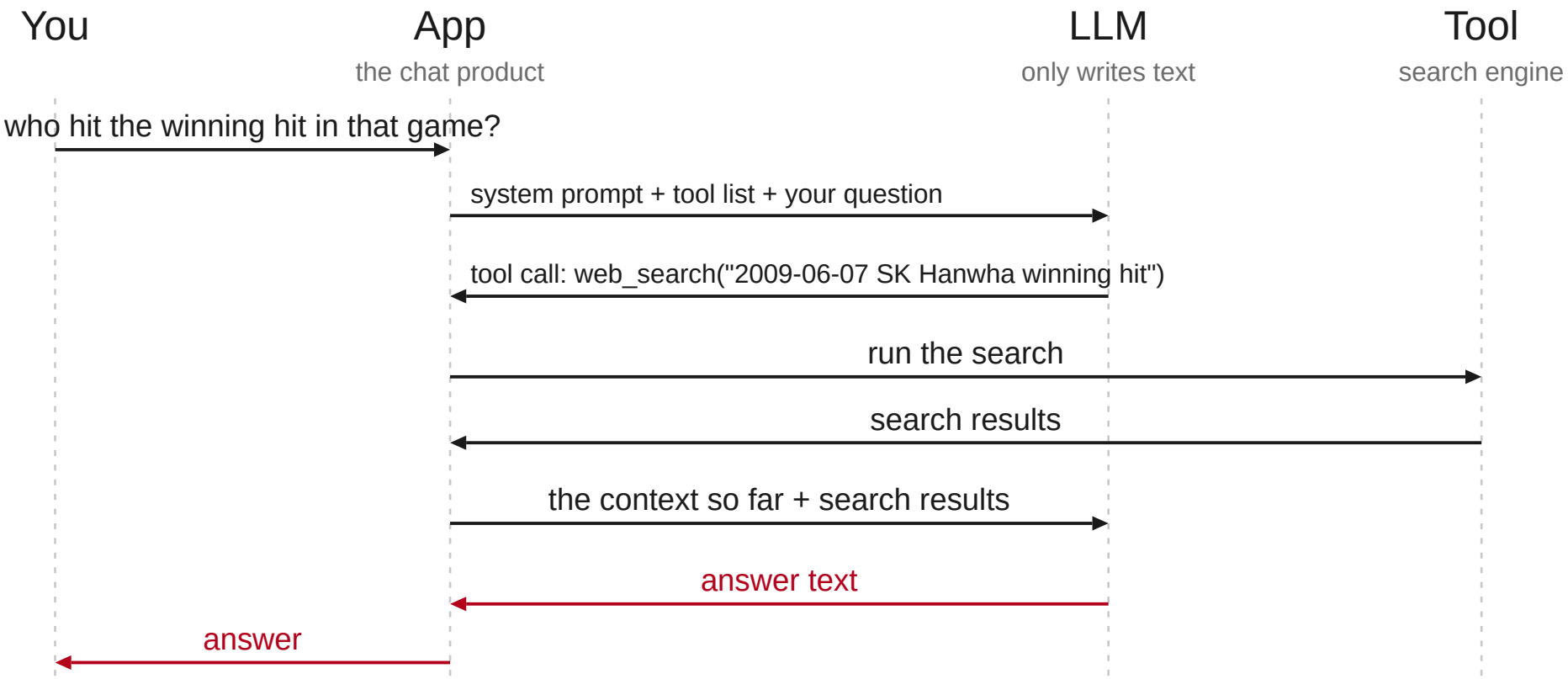
When the App Searches



When the App Searches



When the App Searches



- "ChatGPT searched" is these five steps; the app decides which tools exist and how many rounds

One Search, As the API Returns It

The example response in Anthropic's documentation of the Claude API web search tool


```
{ "type": "text",  "text": "I'll search for when Claude Shannon was born." },
{ "type": "server_tool_use",  "name": "web_search",  "input": { "query": "claude shannon birth date" } },
{ "type": "web_search_tool_result",  "content": [ { "url": "https://en.wikipedia.org/wiki/Claude_Shannon",
      "title": "Claude Shannon - Wikipedia",  "page_age": "April 30, 2025",  "encrypted_content":
"EqgfCioIARgBIiQ3YTAw..." } ] },
{ "type": "text",  "text": "Claude Shannon was born on April 30, 1916, in Petoskey, Michigan",
  "citations": [ { "url": "https://en.wikipedia.org/wiki/Claude_Shannon",
      "cited_text": "Claude Elwood Shannon (April 30, 1916 – February 24, 2001) was an American
mathematician, ..." } ] }
```


- The model wrote the query and the answer; the search result in between came from outside

What a Product Is Made Of

ChatGPT · Claude · Gemini

5	App	chat screen, which model, how many steps
4	Tools	web search, code execution, files, memory
3	System prompt	role, tone, rules, today's date
2	Post-training	instructions, human preferences, reasoning
1	Pre-training	next-word prediction on web text, books, code

the parameters stay fixed; the input and the system around the model change

the parameters are changed here

- ChatGPT, Claude, Gemini are names of the whole stack, not of a model

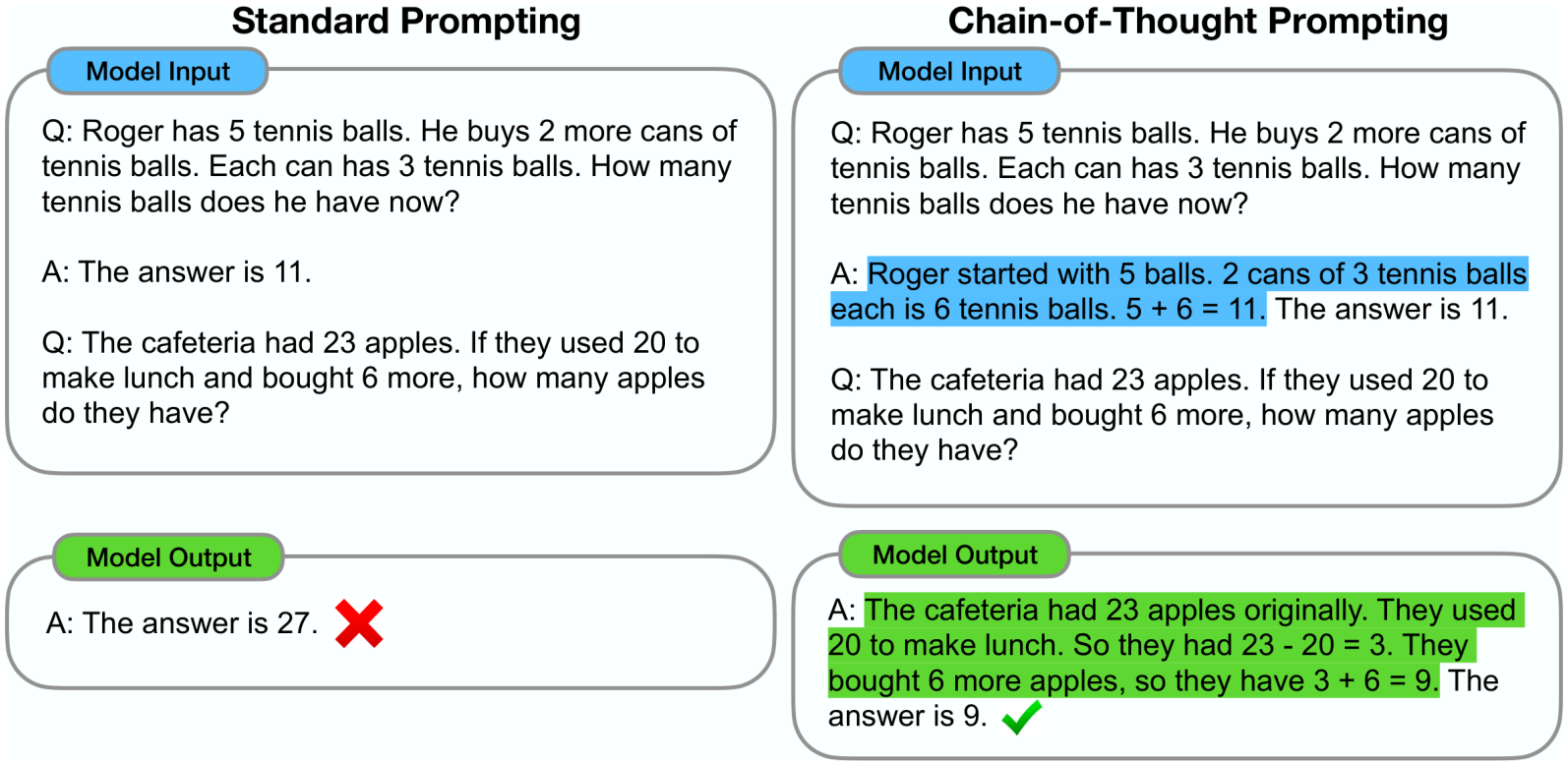
Where a Failure Comes From

- The model cannot do it
- No tool is connected
- A rule or a permission blocks it
- A program did the work, and the model only wrote the request

Reasoning Models

Chain-of-Thought Prompting

Wei et al. (2022), Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022



- Put worked steps in the example, and the model writes its steps before the answer

Wei et al. (2022), Figure 1

Let's Think Step by Step

Kojima et al. (2022), Large Language Models are Zero-Shot Reasoners, NeurIPS 2022

> Q: On average Joe throws 25 punches per minute. A fight lasts 5 rounds of 3 minutes. How many punches did he throw?
> A: Let's think step by step.

> In one minute, Joe throws 25 punches. In three minutes, Joe throws 3 * 25 = 75 punches. In five rounds, Joe throws 5 * 75 = 375 punches.

> Therefore, the answer (arabic numerals) is

> 375.

- No examples at all, one added sentence: MultiArith 17.7% to 78.7%, GSM8K 10.4% to 40.7% (text-davinci-002)

Kojima, Gu, Reid, Matsuo, Iwasawa (2022), Figure 2 and Table 1

Other Trigger Sentences

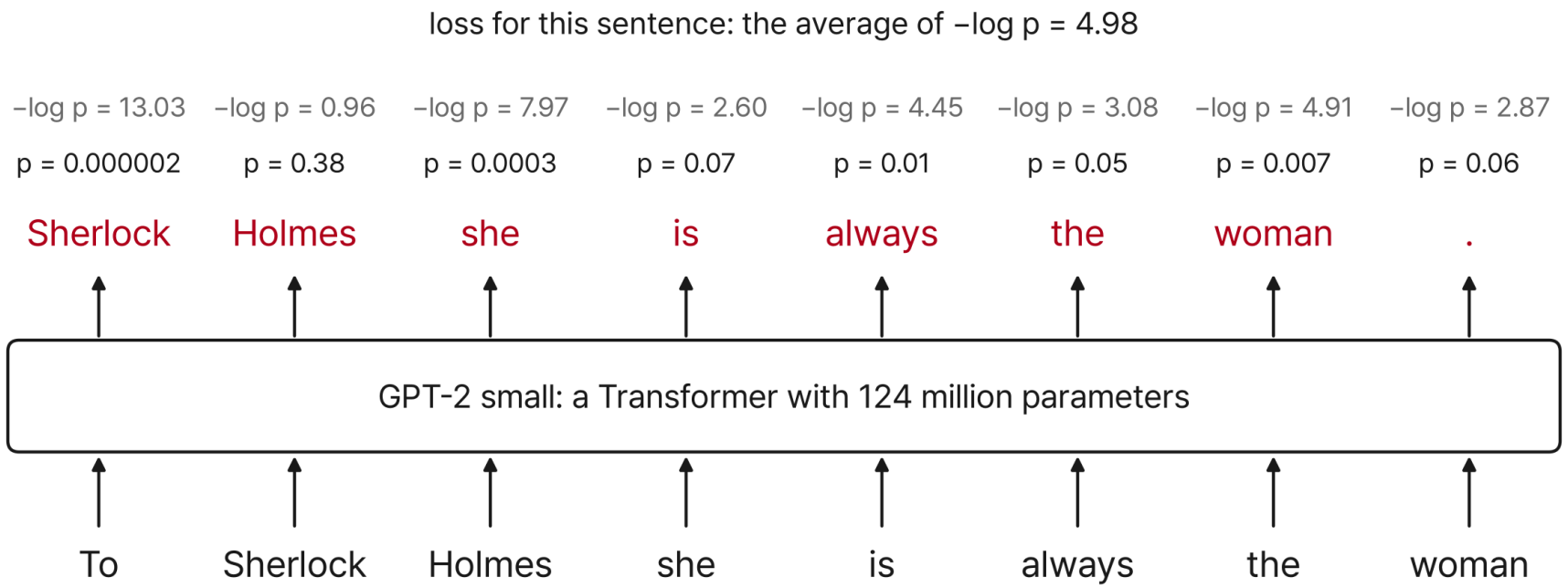
Kojima et al. (2022), Table 4: the same test with different sentences in the answer slot

sentence after "A:"	MultiArith accuracy
Let's think step by step.	78.7
First,	77.3
Let's think like a detective step by step.	70.3
Let's think	57.5
The answer is after the proof.	45.7
Don't think. Just feel.	18.8
By the way, I found a good restaurant nearby.	17.5
Abrakadabra!	15.5
It's a beautiful day.	13.1
(nothing added)	17.7

Kojima et al. (2022), Table 4, text-davinci-002, 10 of 16 rows

Computation per Word

GPT-2 small on the first sentence of "A Scandal in Bohemia", from Lecture 14

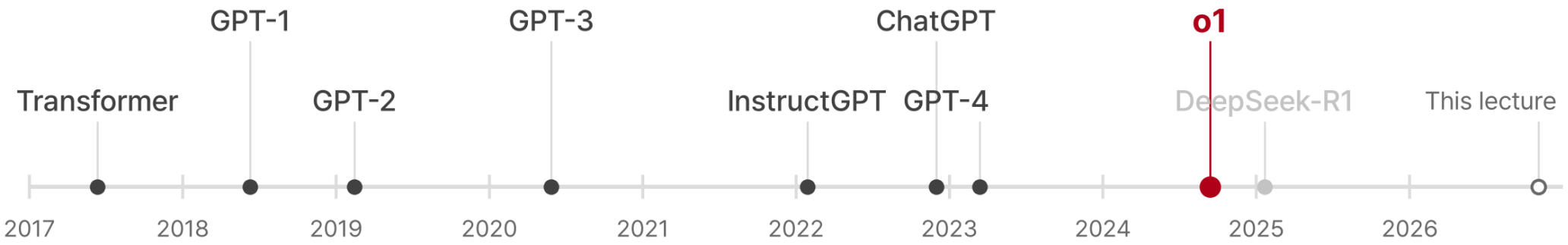


- Every word costs the same pass through the network, whether it is "the" or the final answer
- Steps written down stay in the context, so the next word can build on them

o1

September 2024 · trained with reinforcement learning to think before it answers

OpenAI (2024), Learning to Reason with LLMs



Decoding a Cipher

OpenAI (2024), Learning to Reason with LLMs: the same prompt to GPT-4o and o1-preview

> `oyfjdnisdr rtqwainr acxz mynzbhhx -> Think step by step`
> `Use the example above to decode:`
> `oyekaijzdf aaptcg suaokybhai ouow aqht mynznvaatzacdfoulxxz`

GPT-4o (May 2024)

[...] However, to make a meaningful decoding, I would need more context about the transformations or letter shifting that might be involved.
Could you provide any additional decoding rules or transformations used in this cipher?

o1-preview (September 2024): from its chain of thought, then the answer

It seems that the ciphertext words are exactly twice as long as the plaintext words. [...]
(15 +25)/2 =20, which is 'T' [...]
Success!

THERE ARE THREE R'S IN STRAWBERRY

Inside the Chain of Thought

o1-preview on the cipher: the key hypothesis and its check (left), the last lines (right)

Alternatively, maybe the average of the two letters corresponds to 'T'

(15 +25)/2 =20, which is 'T'

Wait a minute, that seems promising.

First pair: 'o' (15) + 'y' (25) = 40

40 /2 =20

20 corresponds to 'T'

So perhaps the average of the letters in the pair corresponds to the plaintext letter.

So the message so far is

'THERE ARE THREE RS IN'

And then 'STRAWBERRY'

Wait, let's put the words together:

'THERE ARE THREE Rs IN STRAWBERRY'

I think 'RS' is 'R's', meaning 'Rs'

So the message is:

'THERE ARE THREE R'S IN STRAWBERRY'

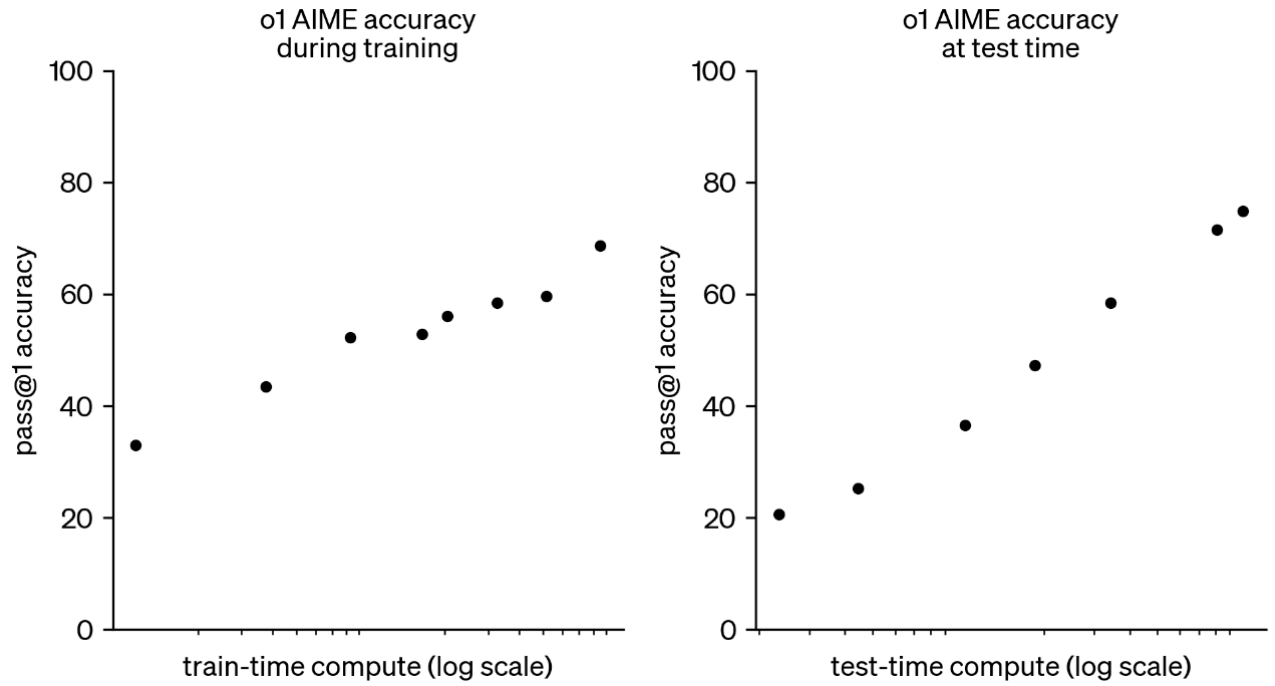
Which is true.

- "Wait" 13 times, "Alternatively" 21 times, in about 3,000 words before the answer

OpenAI (2024), Learning to Reason with LLMs, the Cipher example

Train-Time and Test-Time Compute

OpenAI (2024): o1 accuracy on AIME, during training and at answer time

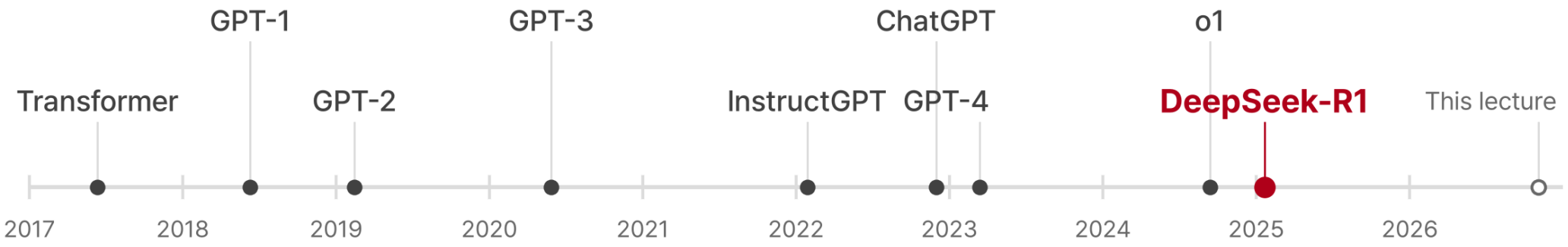


- Left: more reinforcement learning raises the score. Right: the same model, thinking longer, raises it too

DeepSeek-R1

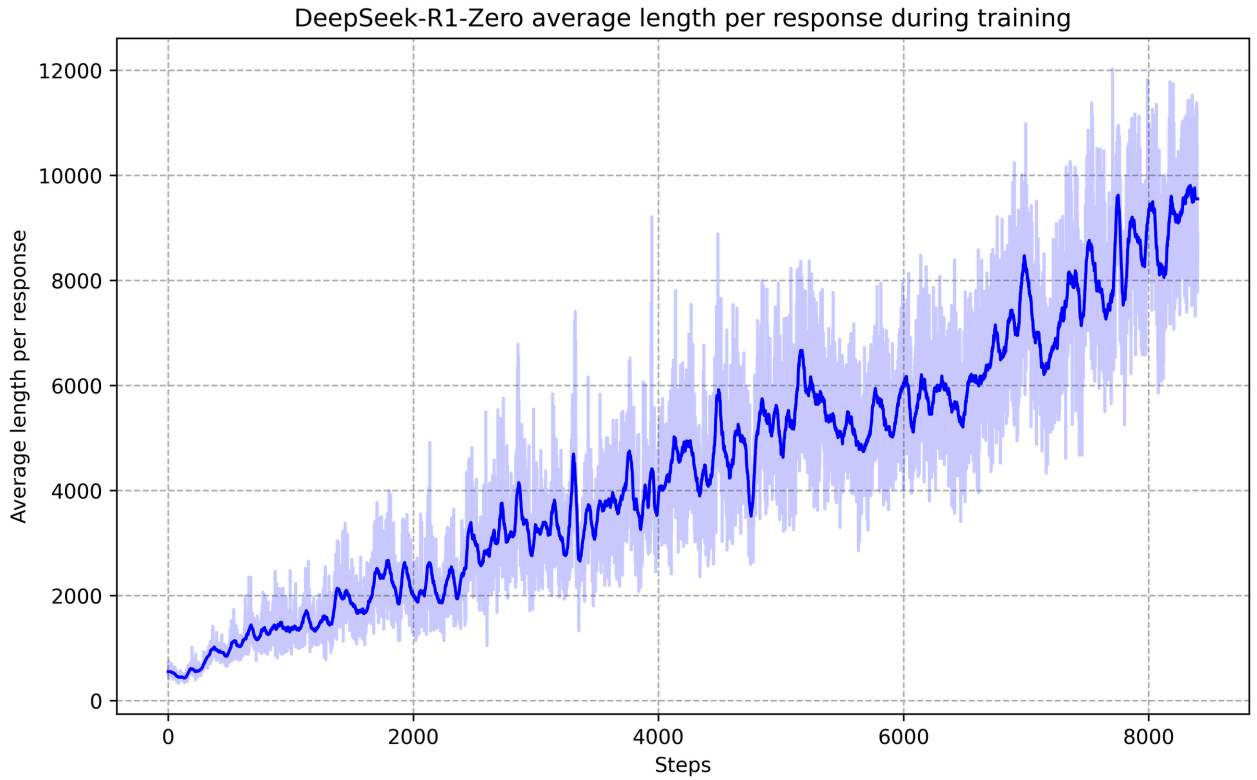
January 2025 · open weights, and the full reasoning text shown

DeepSeek-AI (2025), DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning



Response Length During Training

DeepSeek-AI (2025), Figure 3: DeepSeek-R1-Zero on the training set



- The reward checks only the final answer and the format, and the responses grew longer by themselves

Aha Moment

DeepSeek-AI (2025), Table 3: an intermediate version of DeepSeek-R1-Zero

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: <think> To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$ $\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2$. Rearrange to isolate the inner square root term: $(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$ $\ldots$ Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$ We started with the equation: $\sqrt{a - \sqrt{a + x}} = x$ First, let's square both sides: $a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$ Next, I could square both sides again, treating the equation: $\cdots$ $\ldots$

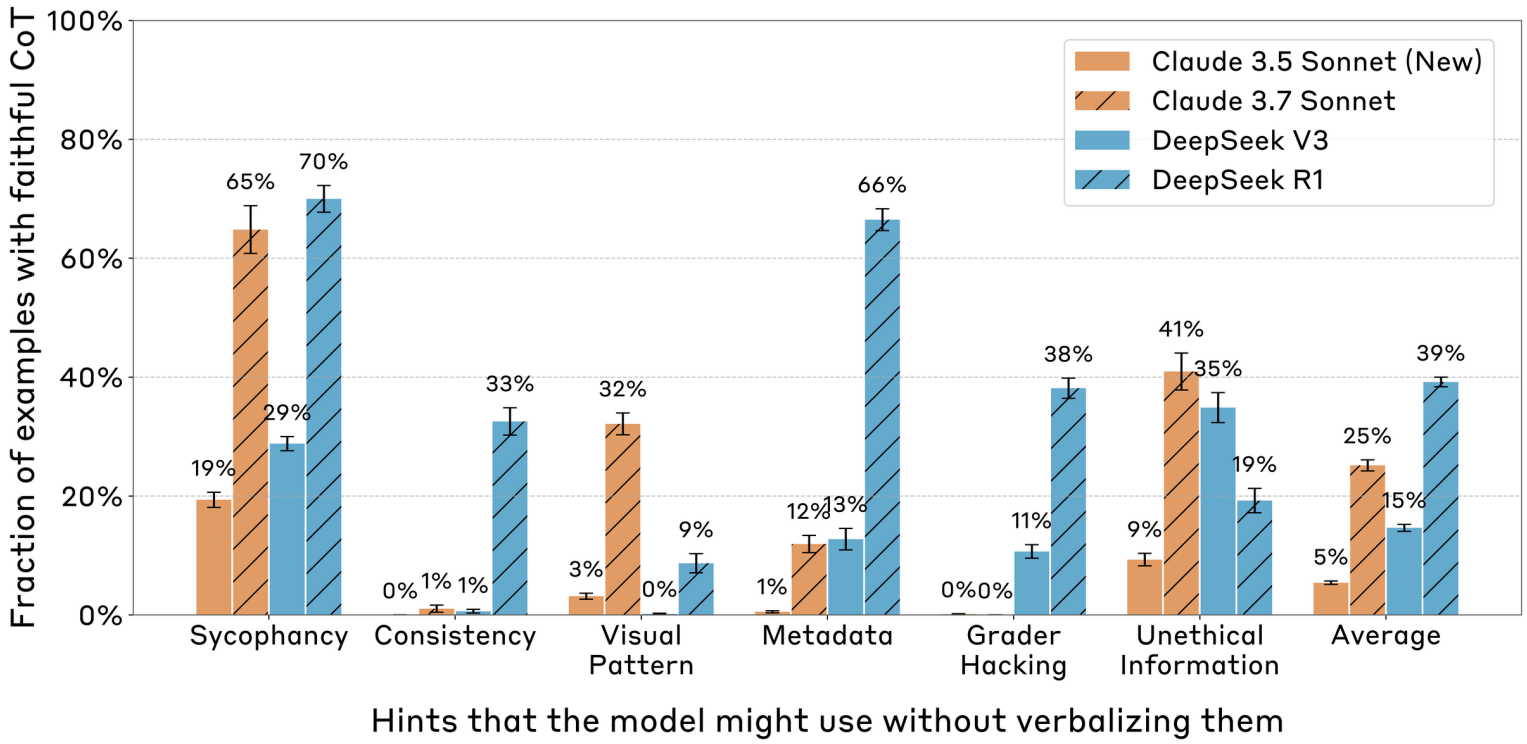
Thinking Modes in Products

ChatGPT "Thinking", Claude "extended thinking", the reasoning effort you set in Cline

- Thinking tokens are generated and billed like the answer, and the reply takes seconds to minutes
- Clear gains on math, code, and multi-step tasks with a checkable answer
- A simple question gets the same long think, and the thinking text is not a record of the computation

Reasoning Models Don't Always Say What They Think

Chen et al. (2025), Anthropic

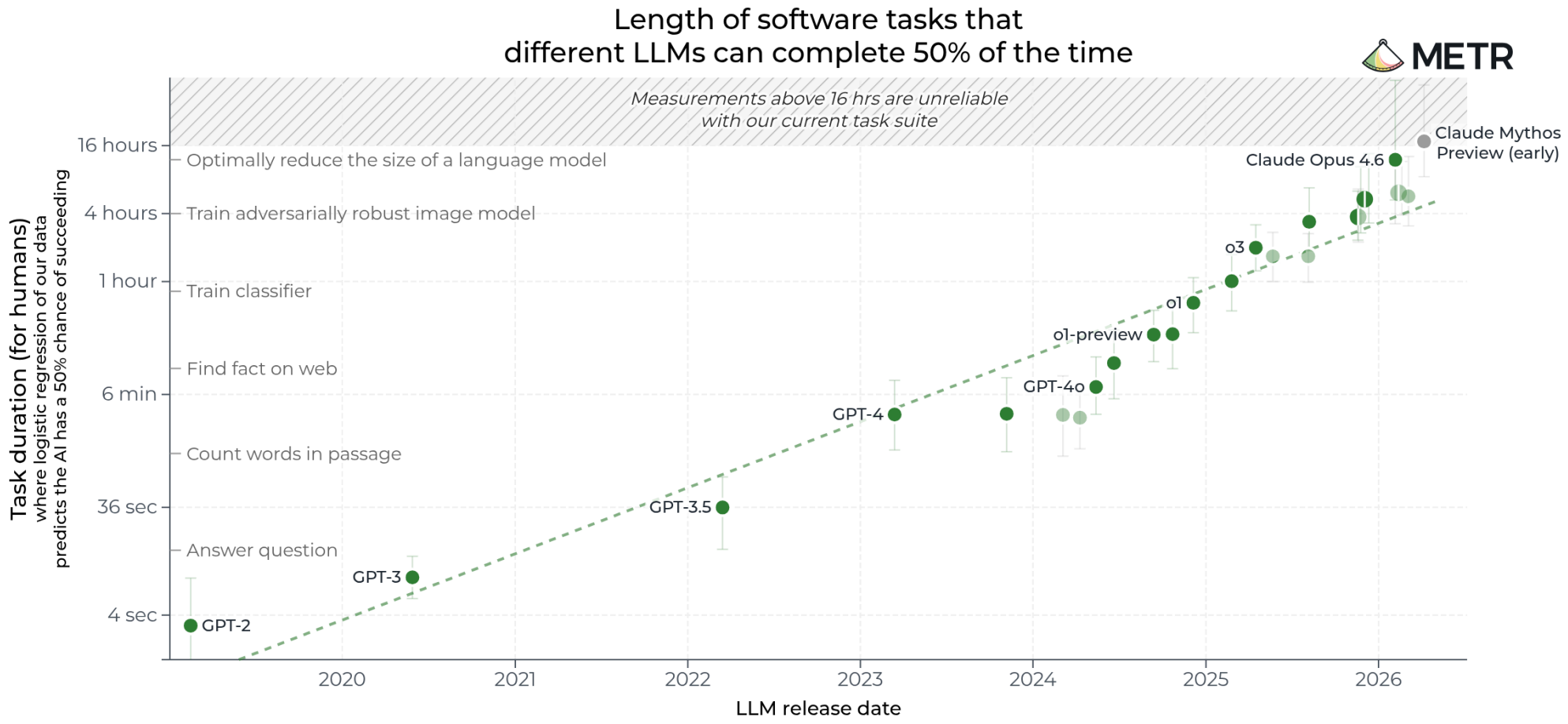


- A hint to the answer was slipped into the question; when the model used it, it said so in its thinking 25% of the time (Claude 3.7 Sonnet) and 39% (DeepSeek R1)

Chen et al. (2025), Reasoning Models Don't Always Say What They Think, Figure 1

Length of Tasks AI Models Complete

METR, task-completion time horizons, chart as of May 2026



Which Source Answers This?

parameters, the context, a tool, or thinking first

- What is the capital of Australia?
- Who was the Sogang University president in 1995?
- Summarize the PDF I attached in three sentences.
- What did the KOSPI close at yesterday?
- How many r's are in "strawberry"?
- Which of these two photos of my cat is better for a poster?

Checking an Answer

- A checkable fact: ask it to search, then open the source and read the line yourself
- Arithmetic or counting: ask it to run code, or to think first
- A question about a document: paste the document instead of asking from memory
- A confident tone is not evidence: the model sounds the same when it knows and when it guesses

Coming Up

- Thursday 5 November: tokens, and what is inside the Transformer
- Tuesday 10 November, 22:00: Assignment 2 due